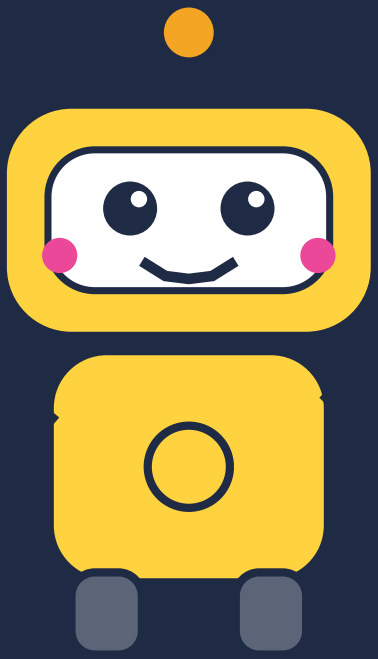
TEACHER BOOK

Think Like a Coder

Teacher book for Class 3

A 5-period plan for every chapter
Hands-on activities, common mistakes and exit checks
Answer keys, assessment guide and photocopiable journal



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 3 · Follows the 14 chapters of NCERT Maths Mela, Class 3

SAMPLE EDITION · DRAFT

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What's inside

About this teacher book . . . 3
How a lesson runs . . . 4
Year plan . . . 5
Assessment . . . 6
1. What's in a Name? . . . 7
2. Toy Joy . . . 9
3. Double Century . . . 11
4. Vacation with My Nani Maa . . . 13
5. Fun with Shapes . . . 15
6. House of Hundreds - I . . . 17
7. Raksha Bandhan . . . 19
8. Fair Share . . . 21
9. House of Hundreds - II . . . 23
10. Fun at Class Party! . . . 25
11. Filling and Lifting . . . 27
12. Give and Take . . . 29
13. Time Goes On . . . 31
14. The Surajkund Fair . . . 33
Observation journal (photocopy) . . . 35
Term summary (photocopy) . . . 36

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About this teacher book

This book goes with the *Think Like a Coder* Class 3 workbook. It gives you a plan for every chapter, so that a maths or class teacher can teach computational thinking confidently without being a computer specialist. CBSE suggests about **50 hours** a year for Class 3; the plans here use **75 periods of 40 minutes**, which is about 50 hours.

★ What is in this book

- ✓ **A year plan** showing which chapter to teach in which term, next to the NCERT maths chapter.
- ✓ **A 5-period plan for every chapter**: what you do and what students do in each period.
- ✓ **A hands-on activity** for every chapter, with set-up steps and a tip.
- ✓ **Common mistakes** to watch for, and how to support or stretch students.
- ✓ **An exit check** with answers, and what to note in the observation journal.
- ✓ **The answer key** for every page of the student workbook.
- ✓ **Photocopiable** observation journal and term summary pages at the back.

✓ Before you start

- ✓ Read the chapter in the student book and these notes. Most lessons need 5 to 10 minutes to prepare.
- ✓ Get the kit items ready (listed under *At a glance*). Everything is reusable.
- ✓ Plan pairs in advance: mix confident readers with children who prefer talking and acting out.
- ✓ Teach the CT chapter in the same week or fortnight as the matching NCERT maths chapter, so the maths is fresh.
- ✓ Keep it active: children should talk, act out and build more than they listen.

No computer lab needed

No computers are needed. Every idea is taught with paper, cards, the floor mat and talk. These are the same ideas children will later use in block coding and AI: steps in order, loops, decisions, data and testing. A projector helps for showing pages but is optional.

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How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.

2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.

3. **Modify** 8 min
   Children change the steps to do something new.

4. **Put in order** 8 min
   Jumbled code cards go back in the right order.

5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

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Year plan

14 chapters of 5 periods each, plus 5 periods for revision and assessment: **75 periods** of 40 minutes in all. Teach each chapter alongside the NCERT maths chapter with the same name; the order and timing can follow your school's maths plan.

Ch	Chapter	Coder skill	Level	Periods	Term
1	What's in a Name?	Algorithms and secret codes	B1	5	Term 1
2	Toy Joy	Abstraction: seeing what matters	B1	5	Term 1
3	Double Century	Patterns and loops	B1	5	Term 1
4	Vacation with My Nani Maa	Breaking problems into steps	B1	5	Term 1
5	Fun with Shapes	Shape patterns and repeat	B2	5	Term 1
6	House of Hundreds - I	Events: when this happens, do that	B2	5	Term 1
7	Raksha Bandhan	Equal groups and sorting data	B2	5	Term 1
8	Fair Share	Rules for sorting and sharing	B2	5	Term 2
9	House of Hundreds - II	Clues and logical thinking	B2	5	Term 2
10	Fun at Class Party!	Sorting: putting things in order	B2	5	Term 2
11	Filling and Lifting	If-then rules and comparing	B3	5	Term 2
12	Give and Take	Variables: keeping count	B2	5	Term 2
13	Time Goes On	Sequences and loops in time	B2	5	Term 2
14	The Surajkund Fair	Grids and robot routes	B1-B2	5	Term 2
	Revision and term assessment	Review games, the term summary and parent feedback	All	5	Both

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Assessment and the observation journal

CBSE asks for continuous, competency-based assessment in CT&AI: written puzzles, group activities and a **Teacher Observation Journal**. This book makes that simple. In each chapter, the exit check and the 'what to note' box tell you what to look for; record one level per child per skill every few chapters, not every lesson.

Skill	Beginning	Developing	Secure	Extending
Following steps	Follows short steps with help	Follows and fixes simple steps	Writes steps with loops	Designs and improves own steps
Patterns	Spots simple repeats	Continues a pattern	Finds the core and rule	Makes new patterns and explains them
Breaking problems down	Needs help to start	Breaks familiar jobs into steps	Breaks new problems into steps	Plans steps and their order alone
Logic and rules	Uses one clue	Uses two clues or an if-then rule	Uses several clues and if-else	Spots extra clues and explains

✓ How to assess without extra tests

- ✓ **Watch and listen** during activities: explaining, predicting and fixing show understanding better than a written answer.
- ✓ **Use the exit check** at the end of each chapter: two quick questions, answers in this book.
- ✓ **Compare with self-checks**: each chapter ends with 'How did I do?', where children colour a face for each 'I can' statement.
- ✓ **Summarise each term** on the term summary page: one level per area and a short comment for the report card.

The four levels

Use the four levels for every skill. **Beginning**: needs help to start. **Developing**: succeeds with prompts or a partner. **Secure**: succeeds alone on familiar problems. **Extending**: solves new problems and explains why. Most children will be Developing or Secure; that is expected and healthy.

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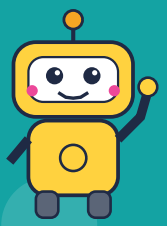
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TEACHER NOTES · CHAPTER 1 · LEVEL B1

What's in a Name?

Coder skill: Algorithms and secret codes

Goes with NCERT Maths Mela, Class 3, chapter 1



At a glance	
Students will	follow steps in order, code and decode a word and find the key of a code.
Key words	**algorithm** (steps done in order), **code** (a hidden message), **key** (the secret number)
Materials	Cipher wheels (kit), alphabet strips, small word cards, pencils
Time	5 periods of 40 minutes
Link	Teach alongside chapter 1 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *algorithm*, *code*, *key*. Work through the example *Code the name RAM with key +1* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Spy messages	Run the activity below. Move between groups and note who can explain the idea.	Code a word with a secret key, swap, and crack another pair's key by trying keys in order.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	These steps code any word with key +2, but they are jumbled. Number them in the right order. Model the first card or step, then let pairs finish.	Now use your steps to code your own name with key +2.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Spy messages

1. Give each pair a cipher wheel and a secret key from +1 to +3. They keep the key hidden.
2. Each pair writes a short word (3 or 4 letters) and codes it, letter by letter.
3. Pairs swap coded words. The other pair tries key +1, then +2, then +3 until the word makes sense.
4. Ask: 'What steps did you follow every time?' Write them on the board as numbered steps: that list is an algorithm.

Tip: Choose words without letters near Z at first. Introduce 'wrapping round' (Y + 2 = A) once children are confident.

Common mistakes to watch for

- **Shifting the wrong way when decoding.** To decode, move *back* by the key.
- **Forgetting to wrap round after Z.** Show the alphabet as a circle: after Z comes A again.
- **Changing the key halfway through a word.** Every letter uses the same key.

Different paces

Support: Use key +1 only, with an alphabet strip instead of the wheel.
Stretch: Crack a longer message without being told the key, and explain the quickest way to find it.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Code CAT with key +1. (Answer: DBU)

2. A word coded with key +2 is ECV. What was the word? (Answer: CAT)

Note in the journal: Beginning (needs help to follow steps in order); **Developing** (can follow steps in order with prompts or a partner); **Secure** (can follow steps in order and code and decode a word alone); **Extending** (can find the key of a code and explain why).

Answer key

1. CBU
2. BOOK
3. a) +1
4. Ali, Tara, Meera, Joseph
5. b) I
Build it: Take the next letter; Move it 2 letters forward; Write the new letter down; Repeat until no letters are left
Challenge: Key +3; FDW = CAT

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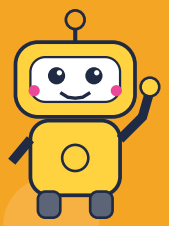
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TEACHER NOTES · CHAPTER 2 · LEVEL B1

Toy Joy

Coder skill: Abstraction: seeing what matters

Goes with NCERT Maths Mela, Class 3, chapter 2



At a glance	
Students will	see a toy from different sides, count blocks in a wall and keep only the details that matter.
Key words	**view** (how a thing looks from one side), **abstraction** (keeping only what matters)
Materials	Building blocks (or the kit picture cards), a tin or drum, a book to hide the tower behind, paper
Time	5 periods of 40 minutes
Link	Teach alongside chapter 2 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *view*, *abstraction*. Work through the example *A drum from two sides* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Blind builder	Run the activity below. Move between groups and note who can explain the idea.	Give and follow building instructions without seeing the model, then improve the unclear steps.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write instruction cards so a friend can build this tower without seeing it. Model the first card or step, then let pairs finish.	Swap cards with a friend. Do their instructions build the same tower?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Blind builder

1. In pairs, child A builds a tower of 4 to 6 blocks behind a book, where child B cannot see it.
2. A gives spoken instructions, one step at a time. B builds without looking at A's tower.
3. Remove the book and compare. Find the instruction that was unclear and rewrite it together.
4. Swap roles. Ask: 'Which details mattered, and which did not?' (Position matters; the colour of the table does not.)

Tip: Allow words only, no pointing. That is what makes children choose the important details.

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Common mistakes to watch for

- **Describing details that do not matter** and missing the ones that do, like 'on top of' or 'next to'.
- **Mixing up the top view and the side view.** Let children look at a tin from above and from the side.
- **Instructions that need the listener to see**, such as 'put it there'.

Different paces

Support: Build towers of only 3 blocks in a straight line.
Stretch: Write instructions that use no more than 6 words per step, and test them with a new partner.

Exit check and what to note

Two quick questions at the end of the chapter:

1. What shape is a tin when you look at it from the top? (Answer: a circle)

2. Name one detail that does NOT matter when describing a tower to build. (Answer: for example the colour of the table, or who owns the blocks)

Note in the journal: Beginning (needs help to see a toy from different sides); **Developing** (can see a toy from different sides with prompts or a partner); **Secure** (can see a toy from different sides and count blocks in a wall alone); **Extending** (can keep only the details that matter and explain why).

Answer key

1. 1 ball: B; 2 dice: A; 3 party cap: C
2. 8
3. b) the roads
4. b
5. 10 blocks (1 + 2 + 3 + 4)
Build it: Example: Put a blue block. Put an orange block on top of it. Put an orange block next to the blue block.
Challenge: 14 blocks; the 4th layer has 16 (4 x 4)

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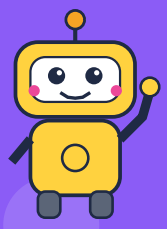
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TEACHER NOTES · CHAPTER 3 · LEVEL B1

Double Century

Coder skill: Patterns and loops

Goes with NCERT Maths Mela, Class 3, chapter 3



At a glance	
Students will	continue a number pattern, find the step in a loop and write my own loop card.
Key words	**loop** (repeat the same step), **pattern** (things that follow a rule)
Materials	Chalk or tape for a floor number line (0 to 300, marked in tens), loop cards, counters
Time	5 periods of 40 minutes
Link	Teach alongside chapter 3 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *loop*, *pattern*. Work through the example *Start at 100. Repeat 3 times: add 50.* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Jumping on the number line	Run the activity below. Move between groups and note who can explain the idea.	Predict where a loop lands, act it out on the number line, then design their own loops.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Make a loop card that goes from 100 to 200. Fill in the blanks. Model the first card or step, then let pairs finish.	Can you find two different loops that both reach 200?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Jumping on the number line

1. Mark a number line on the floor from 0 to 300, with a mark every 10 and big marks at every 50.
2. A child stands on 100. Show the loop card *repeat 2 times: jump +50*. The class predicts the landing number first.
3. The child jumps while the class counts each jump aloud. Check the prediction.
4. Ask pairs to find other loops from 100 that land on 200, such as *repeat 10 times: +10* or *repeat 4 times: +25*.

Tip: Make children say the loop aloud before jumping: 'repeat, jump, repeat, jump'. It helps them see the loop as one block.

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Common mistakes to watch for

- **Counting the starting number as the first jump.** Ask: 'Have you moved yet?'
- **Repeating one time too many or too few.** Tick the loop card each time a jump is made.
- **Adding the repeat number instead of the step.** *Repeat 3 times: +50* adds 150, not 3.

Different paces

Support: Use steps of 10 on a 100-square before moving to the floor line.
Stretch: Find a loop from 100 to 200 whose step is not a multiple of 10, and one that goes backwards from 300.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Start at 100. Repeat 4 times: add 25. Where do you end? (Answer: 200)

2. Start at 50. Which loop reaches 200: repeat 3 times +50, or repeat 5 times +50? (Answer: repeat 3 times +50)

Note in the journal: Beginning (needs help to continue a number pattern); **Developing** (can continue a number pattern with prompts or a partner); **Secure** (can continue a number pattern and find the step in a loop alone); **Extending** (can write my own loop card and explain why).

Answer key

1. 150, 160
2. b) add 5
3. 300
4. c) 135
5. 5 hops
Build it: Any loop that adds up to 100: e.g. start 100, repeat 2 times add 50; or repeat 4 times add 25; or repeat 10 times add 10.
Challenge: After 10 steps both show 200: a double century!

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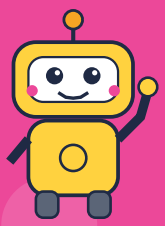
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TEACHER NOTES · CHAPTER 4 · LEVEL B1

Vacation with My Nani Maa

Coder skill: Breaking problems into steps

Goes with NCERT Maths Mela, Class 3, chapter 4



At a glance	
Students will	break a big job into steps, put steps in order and find and fix a bug.
Key words	**decompose** (break into small steps), **bug** (a mistake in the steps), **debug** (find and fix it)
Materials	Step cards, a plate, bread and butter pictures (or real items), the trip map from the book
Time	5 periods of 40 minutes
Link	Teach alongside chapter 4 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *decompose*, *bug*, *debug*. Work through the example *The trip to Nani Maa's house* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The sandwich robot	Run the activity below. Move between groups and note who can explain the idea.	Give the robot steps, spot each bug, and rewrite the steps in a working order.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Robo wrote a plan to water the plant, but it has a bug. Find it and fix it. Model the first card or step, then let pairs finish.	Write the fixed order of the four cards.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The sandwich robot

1. You are a robot that follows instructions *exactly*. Ask the class to tell you how to make a pretend sandwich.
2. Follow every instruction literally. 'Put butter on the bread' might mean putting the whole butter dish on the bread!
3. Each time something goes wrong, the class finds the bug: a missing step, a step in the wrong order, or a step that is too big.
4. Write the fixed steps on the board, then break one big step ('get ready for school') into smaller ones together.

Tip: Exaggerate the literal mistakes. Children love fixing the robot, and they remember why order and detail matter.

Common mistakes to watch for

- **Skipping 'obvious' steps**, like opening the packet before taking the bread.
- **Putting a step before the thing it needs**, like eating before making the sandwich.
- **Writing one giant step** ('make the sandwich') instead of breaking it down.

Different paces

Support: Order 4 picture cards instead of writing steps.
Stretch: Break 'get ready for school' into at least 6 steps in the right order, and check them with a partner acting as the robot.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Put in order: B) put on socks, A) put on shoes, C) tie the laces. (Answer: B, A, C)

2. Find the bug: 1. Drink the water. 2. Take a glass. 3. Fill it with water. (Answer: drinking must come last: 2, 3, 1)

Note in the journal: Beginning (needs help to break a big job into steps); **Developing** (can break a big job into steps with prompts or a partner); **Secure** (can break a big job into steps and put steps in order alone); **Extending** (can find and fix a bug and explain why).

Answer key

1. Example order: 1 cut a lemon, 2 pour water in a glass, 3 squeeze the lemon in, 4 add sugar, 5 stir it
2. a) 1 and 2
3. Rs 160
4. c) 12 o'clock
5. Example: wake up, brush teeth, bathe and dress, pack the school bag
Build it: Fill the can; walk to the plant; pour water on the plant; put the can back (steps 2 and 3 were swapped)
Challenge: 2. Take two slices of bread; 4. Spread butter; 3. Put the cucumber on top; 1. Eat the sandwich

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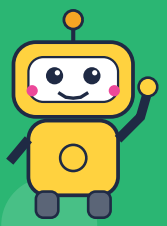
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TEACHER NOTES · CHAPTER 5 · LEVEL B2

Fun with Shapes

Coder skill: Shape patterns and repeat

Goes with NCERT Maths Mela, Class 3, chapter 5



At a glance	
Students will	find the core of a pattern, continue a growing pattern and draw from a repeat program.
Key words	**pattern** (follows a rule), **core** (the part that repeats), **growing** (gets bigger each step)
Materials	Shape cards (kit), paper shapes or beads, a space on the floor or a table
Time	5 periods of 40 minutes
Link	Teach alongside chapter 5 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *pattern*, *core*, *growing*. Work through the example *Find the core* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Pattern trains	Run the activity below. Move between groups and note who can explain the idea.	Build a repeating pattern, find its core, write it as a repeat program and predict a later shape.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Follow this program and draw what it makes in the space below. Model the first card or step, then let pairs finish.	Now write your own repeat program for a friend to draw.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Pattern trains

1. Groups make a 'train' of shape cards that repeats, for example triangle, circle, circle, triangle, circle, circle.
2. Another group finds the **core**, the smallest part that repeats, and writes it as a program: *repeat 3 times: triangle, circle, circle*.
3. Without looking further, they predict the 10th shape. Then extend the train to check.
4. End with a growing pattern (2, 4, 6 blocks) and ask: 'Does this repeat, or grow? What is its rule?'

Tip: Ask children to clap the pattern as they say it. The beat makes the core easy to hear.

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Common mistakes to watch for

- **Choosing a core that is too long or too short.** The core is the smallest part that repeats.
- **Counting from 0** when predicting a position. The first shape is position 1.
- **Thinking every pattern repeats.** Some patterns grow instead, by adding the same amount each step.

Different paces

Support: Start with a two-shape core, such as red, blue.
Stretch: Make a growing pattern, write its rule, and predict step 10 without drawing.

Exit check and what to note

Two quick questions at the end of the chapter:

1. The core is star, circle, circle. What is the 7th shape? (Answer: star (stars are at positions 1, 4 and 7))

2. Step 1 has 3 blocks, step 2 has 6, step 3 has 9. How many blocks are in step 5? (Answer: 15)

Note in the journal: Beginning (needs help to find the core of a pattern); **Developing** (can find the core of a pattern with prompts or a partner); **Secure** (can find the core of a pattern and continue a growing pattern alone); **Extending** (can draw from a repeat program and explain why).

Answer key

1. star, circle
2. b) square, square, diamond
3. 7
4. c) E
5. a) up
Build it: square, circle, square, circle, square, circle
Challenge: 20 (add 2 each step)

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6

TEACHER NOTES · CHAPTER 6 · LEVEL B2

House of Hundreds - I

Coder skill: Events: when this happens, do that

Goes with NCERT Maths Mela, Class 3, chapter 6



At a glance	
Students will	follow 'when this happens' rules, read hundreds, tens and ones and make the largest number.
Key words	**event** (something that happens), **hundreds, tens, ones** (the places of a number)
Materials	Button cards (+1, +10, +100), place-value blocks or bundles, board for the 'counter'
Time	5 periods of 40 minutes
Link	Teach alongside chapter 6 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *event, hundreds, tens, ones*. Work through the example *The counter machine* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The button game	Run the activity below. Move between groups and note who can explain the idea.	Act as buttons and counter, follow events, and plan the fewest presses to reach a target.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Design your own counter machine. Choose what each button does. Model the first card or step, then let pairs finish.	Start at 500. Ask a friend to press 3 buttons and tell you the number.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: The button game

1. Three children hold the button cards: +1, +10 and +100. One child is the counter and writes the number on the board, starting at 345.
2. Call out 'press +10!' (an **event**). The counter updates the number: when this happens, do that.
3. After a few presses, give a target, such as 456, and ask the class for the fewest presses to reach it.
4. Swap roles. Try a target that needs a carry, such as 395 to 405.

Tip: Use place-value bundles for each press, so children see which digit changes and why.

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Common mistakes to watch for

- **Changing the wrong digit**, such as adding 10 to the ones.
- **Forgetting to carry** when the tens go past 9: 395 + 10 = 405.
- **Mixing up the event and the action.** The event is the press; the action is 'add 10'.

Different paces

Support: Show each press with place-value blocks before writing the number.
Stretch: Add a -10 button. What is the fewest number of presses from 200 to 390? (3: +100, +100, -10)

Exit check and what to note

Two quick questions at the end of the chapter:

1. Start at 245. Press +100, +10, +10. What is the number now? (Answer: 365)

2. What is the fewest presses of +1, +10 and +100 to go from 0 to 213? (Answer: 6: two +100, one +10 and three +1)

Note in the journal: Beginning (needs help to follow 'when this happens' rules); **Developing** (can follow 'when this happens' rules with prompts or a partner); **Secure** (can follow 'when this happens' rules and read hundreds, tens and ones alone); **Extending** (can make the largest number and explain why).

Answer key

1. 230
2. b) 505
3. b) red, red, green, green
4. 234
5. 730
Build it: Open answer
Challenge: 5 presses: red, red, red, green, green

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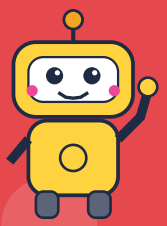
7

TEACHER NOTES · CHAPTER 7 · LEVEL B2

Raksha Bandhan

Coder skill: Equal groups and sorting data

Goes with NCERT Maths Mela, Class 3, chapter 7



At a glance	
Students will	count equal groups with a loop, sort and count data and read a picture graph.
Key words	**equal groups** (same number in each), **data** (facts we collect and count), **sort** (group by a rule)
Materials	Counters or bottle caps, small paper packets or cups, rakhi picture cards, tally chart on the board
Time	5 periods of 40 minutes
Link	Teach alongside chapter 7 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *equal groups*, *data*, *sort*. Work through the example *Rakhis in packets* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Packing packets	Run the activity below. Move between groups and note who can explain the idea.	Pack counters into equal groups, write each as a loop, and sort and tally picture cards.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a loop card for 6 friends who each get 2 rakhis. Model the first card or step, then let pairs finish.	How many rakhis do you need? Show it with a drawing.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Packing packets

1. Give each group 24 counters. Ask them to pack them into equal packets of 2, then of 3, 4 and 6.
2. For each packing, write a loop: *repeat 6 times: put 4 in a packet*.
3. Sort the rakhi picture cards by colour and record the counts with tally marks.
4. Ask: 'Which colour is most common? How do the tally marks help you answer quickly?'

Tip: Cross every fifth tally mark. Counting in fives is faster and shows children why we organise data.

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Common mistakes to watch for

- **Unequal packets** from careless counting. Ask groups to check each packet.
- **Mixing up the number of packets and the number in each packet.** Say both aloud: '6 packets of 4'.
- **Tally marks without the fifth crossing mark**, which makes counting slow and error-prone.

Different paces

Support: Use 12 counters and packets of 2 and 3.
Stretch: Find every way to pack 24 into equal packets with none left over (1, 2, 3, 4, 6, 8, 12 or 24 in each).

Exit check and what to note

Two quick questions at the end of the chapter:

1. There are 5 packets with 4 rakhis in each. How many rakhis? (Answer: 20)

2. 18 sweets are packed in 3s. How many packets? (Answer: 6)

Note in the journal: Beginning (needs help to count equal groups with a loop); **Developing** (can count equal groups with a loop with prompts or a partner); **Secure** (can count equal groups with a loop and sort and count data alone); **Extending** (can read a picture graph and explain why).

Answer key

1. 18
2. c) 4
3. 6
4. b) repeat 3 times: add 5
5. 10
Build it: Repeat 6 times: give 2 rakhis = 12 rakhis
Challenge: 18 (6 rakhis + 12 sweets)

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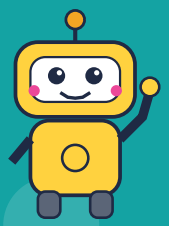
TEACHER NOTES · CHAPTER 8 · LEVEL B2

8

Fair Share

Coder skill: Rules for sorting and sharing

Goes with NCERT Maths Mela, Class 3, chapter 8



At a glance	
Students will	tell if a share is fair, find halves and use an if-else rule.
Key words	**half** (one of 2 equal parts), **fair** (everyone gets the same), **if-else** (a rule with two choices)
Materials	Counters or laddoo cut-outs, two baskets or boxes labelled A and B, fruit picture cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 8 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *half*, *fair*, *if-else*. Work through the example *Sharing 8 laddoos between 2 friends* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The sorting hat	Run the activity below. Move between groups and note who can explain the idea.	Guess a hidden sorting rule, test it, write their own if-else rule, and share counters fairly.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write an if-else rule to sort fruits into two baskets. Model the first card or step, then let pairs finish.	Test your rule with apple, banana and mango. Does every fruit go somewhere?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: The sorting hat

1. Think of a secret rule, for example *if it is yellow, basket A, else basket B*.
2. Children bring fruit cards one at a time; you sort them by your rule without saying it.
3. After a few cards, pairs guess the rule and test it on new cards.
4. Pairs make their own rule and play again. Finish by sharing counters fairly: deal them one by one until none can be shared.

Tip: Include a tricky card (a green apple, a yellow mango) so children see that the rule decides, not what 'feels right'.

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Common mistakes to watch for

- **Rules that leave an item with nowhere to go.** Every if needs an else.
- **Rules where an item could go in both baskets.** Check each item against the rule.
- **Unfair sharing.** Deal one at a time and check that each friend has the same number.

Different paces

Support: Use rules about one property only, such as colour.
Stretch: Make a rule with three baskets: if ..., else if ..., else ....

Exit check and what to note

Two quick questions at the end of the chapter:

1. Share 10 pencils fairly among 3 friends. How many does each get, and how many are left? (Answer: 3 each, 1 left over)

2. Rule: if it is round, box A, else box B. Where does a banana go? (Answer: box B)

Note in the journal: Beginning (needs help to tell if a share is fair); **Developing** (can tell if a share is fair with prompts or a partner); **Secure** (can tell if a share is fair and find halves alone); **Extending** (can use an if-else rule and explain why).

Answer key

1. a) a and c
2. SMALL (50 is not more than 50)
3. Even: 12, 20, 4. Odd: 7, 15
4. 3
5. no: one child gets twice as many
Build it: Example: If the fruit is yellow, basket A; else basket B
Challenge: 2 each, 1 left over; they could break it into 3 equal pieces

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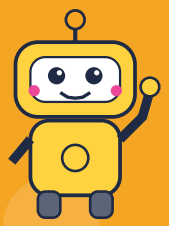
9

TEACHER NOTES · CHAPTER 9 · LEVEL B2

House of Hundreds - II

Coder skill: Clues and logical thinking

Goes with NCERT Maths Mela, Class 3, chapter 9



At a glance	
Students will	use clues to find a number, spot an extra clue and write my own clues.
Key words	**clue** (a hint that rules some answers out), **logic** (thinking step by step to be sure)
Materials	3-digit number cards or a 100-999 number strip, clue cards, small whiteboards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 9 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *clue*, *logic*. Work through the example *Find the mystery number* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Mystery number	Run the activity below. Move between groups and note who can explain the idea.	Use clues one at a time to narrow down a mystery number, then write clues for a partner.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Think of a 3-digit number. Write 3 clues so a friend can find it. Model the first card or step, then let pairs finish.	Can your friend solve it with only two of your clues?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Mystery number

1. Think of a 3-digit number. Give one clue at a time, such as 'I am between 300 and 400'.
2. After each clue, pairs write all the numbers that still fit, or cross out ones that do not.
3. When only one number is left, pairs explain which clue removed the last wrong answers.
4. Pairs write 3 clues for their own number and test them on another pair.

Tip: Ask: 'Could any other number fit all the clues?' This is how children learn to check that clues are enough.

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Common mistakes to watch for

- **Using a new clue but forgetting the earlier ones.** Every answer must fit all the clues.
- **Mixing up the tens digit and the ones digit.** Label the places H, T, O.
- **Writing clues that fit many numbers, or none.** Test your clues before giving them.

Different paces

Support: Use 2-digit numbers and two clues.
Stretch: Write the fewest clues that still lead to only one number.

Exit check and what to note

Two quick questions at the end of the chapter:

1. I am a 3-digit number. My hundreds digit is 4, my tens digit is 2 more than that, and my ones digit is 0. Who am I? (Answer: 460)

2. My digits add up to 5 and I am between 100 and 200. Give two numbers that fit. (Answer: any two of 104, 113, 122, 131, 140)

Note in the journal: Beginning (needs help to use clues to find a number); **Developing** (can use clues to find a number with prompts or a partner); **Secure** (can use clues to find a number and spot an extra clue alone); **Extending** (can write my own clues and explain why).

Answer key

1. 540
2. 555
3. 238
4. d) D, because clue A already says it is less than 200; the number is 163
5. 413, 417, 425
Build it: Open answer
Challenge: 244

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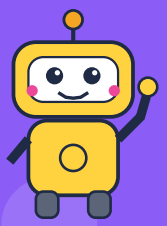
10

TEACHER NOTES · CHAPTER 10 · LEVEL B2

Fun at Class Party!

Coder skill: Sorting: putting things in order

Goes with NCERT Maths Mela, Class 3, chapter 10



At a glance	
Students will	put things in order, compare and swap and write a sorting program.
Key words	**sort** (put in order), **compare** (see which is bigger), **swap** (change places)
Materials	Large number cards (one per child in a line of 5), sorting code cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 10 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *sort*, *compare*, *swap*. Work through the example *Sort 5, 2, 4 from smallest to biggest* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Human sorting	Run the activity below. Move between groups and note who can explain the idea.	Sort a line of number cards using only compare-and-swap, count the swaps, and compare different starting orders.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Put these sorting cards in the right order to make a sorting program. Model the first card or step, then let pairs finish.	Use your program to sort the numbers on 4 cards: 9, 3, 7, 1.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Human sorting

1. Five children hold number cards and stand in a line in a jumbled order.
2. The rule: only neighbours may compare, and they swap if the left number is bigger.
3. The class directs each compare and swap, while one child counts the swaps.
4. Try again with the cards in reverse order. Ask: 'Which start needed the most swaps? Why?'

Tip: Let the children holding cards say 'compare' and 'swap' aloud. Hearing the steps makes the algorithm visible.

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Common mistakes to watch for

- **Swapping cards that are not neighbours.** Only neighbours may swap.
- **Stopping too early.** Go along the line once more to check no swaps are needed.
- **Sorting the wrong way**, biggest first when smallest first was asked.

Different paces

Support: Sort 3 cards first.
Stretch: Sort children by height using only compare-and-swap, without talking.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Sort 8, 3, 5 from smallest to biggest. How many swaps of neighbours does it take? (Answer: 3, 5, 8 in 2 swaps (8 with 3, then 8 with 5))

2. Which kind of list needs no swaps at all? (Answer: a list already in order, like 1, 4, 6)

Note in the journal: Beginning (needs help to put things in order); **Developing** (can put things in order with prompts or a partner); **Secure** (can put things in order and compare and swap alone); **Extending** (can write a sorting program and explain why).

Answer key

1. Ben, Dev, Asha, Chen
2. a) 2, 4, 1, 5
3. b) measuring tape
4. 150 ml, 250 ml, 300 ml, 500 ml
5. 4
Build it: Compare two neighbours; If the first is bigger, swap; Move one place right; Repeat until no swaps. 9, 3, 7, 1 becomes 1, 3, 7, 9
Challenge: 3 swaps: (3,2), (3,1), (2,1)

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TEACHER NOTES · CHAPTER 11 · LEVEL B3

11

Filling and Lifting

Coder skill: If-then rules and comparing

Goes with NCERT Maths Mela, Class 3, chapter 11



At a glance	
Students will	follow an if-then rule, compare weights on a balance and compare how much things hold.
Key words	**if-then** (a rule that makes a decision), **heavier** (weighs more), **capacity** (how much a thing can hold)
Materials	A simple balance (or a ruler on a pencil), 3 or 4 bags of rice or sand of different weights, water bottles
Time	5 periods of 40 minutes
Link	Teach alongside chapter 11 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *if-then*, *heavier*, *capacity*. Work through the example *The tile rule* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Heavier or lighter?	Run the activity below. Move between groups and note who can explain the idea.	Compare bags on a balance, record results as if-then rules, and find the heaviest in the fewest weighings.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write if-then rules for watering a plant. Model the first card or step, then let pairs finish.	What should happen if the pot is already full of water?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: Heavier or lighter?

1. Show the balance. Ask: 'If the left side goes down, what does that tell us?' Write it as a rule: *if left goes down, then left is heavier*.
2. Groups compare two bags, then use the heavier one against the third bag.
3. Ask: 'How many weighings did you need to find the heaviest of 3 bags?' (2)
4. Finish with if-then rules from daily life, such as *if the soil is dry, then water the plant*.

Tip: Keep the heavier bag each time and only compare it with a new bag. That is the 'keep the winner' strategy.

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Common mistakes to watch for

- **Reading the balance backwards.** The side that goes down is heavier.
- **Comparing every pair** instead of keeping the heavier bag, which takes more weighings than needed.
- **If-then rules with no 'otherwise'**, so some cases have no action.

Different paces

Support: Compare two bags only, and say the rule aloud each time.
Stretch: With 4 bags, show that 3 weighings are always enough to find the heaviest.

Exit check and what to note

Two quick questions at the end of the chapter:

1. The left pan of a balance goes down. Which side is heavier? (Answer: the left)

2. How many weighings are needed to find the heaviest of 3 bags? (Answer: 2)

Note in the journal: Beginning (needs help to follow an if-then rule); **Developing** (can follow an if-then rule with prompts or a partner); **Secure** (can follow an if-then rule and compare weights on a balance alone); **Extending** (can compare how much things hold and explain why).

Answer key

1. b) melon
2. 1, 2, 4, 6, 7 then hop 2 to the end
3. b) umbrella
4. a) bucket
5. c) pencil
Build it: Example: If the soil is wet, then do not water
Challenge: 3: compare bags 1 and 2, then the heavier with bag 3, then the heavier with bag 4

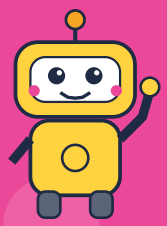
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TEACHER NOTES · CHAPTER 12 · LEVEL B2

12 Give and Take

Coder skill: Variables: keeping count

Goes with NCERT Maths Mela, Class 3, chapter 12



At a glance	
Students will	use a variable to keep count, trace a number step by step and find the start by undoing steps.
Key words	**variable** (a named box for a number), **trace** (write the number after each step)
Materials	A box labelled with a variable name (such as *marbles*), counters, small whiteboards for trace tables
Time	5 periods of 40 minutes
Link	Teach alongside chapter 12 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *variable*, *trace*. Work through the example *Trace Sonu's marbles* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The variable box	Run the activity below. Move between groups and note who can explain the idea.	Change a 'variable box' as a program runs, record a trace table, and write programs for a partner.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a program for a game score. Use a variable called **score**. Model the first card or step, then let pairs finish.	Play: catch, catch, drop, catch. Trace the score.
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: The variable box

1. Label a box *marbles* and put 10 counters inside: 'Set marbles to 10.'
2. Read a short program: *give away 3, get 5, give away 4*. A child changes the counters in the box at each step.
3. The class records each step in a trace table: step, what happens, marbles.
4. Pairs write a 4-step program for each other and trace it.

Tip: Insist on writing the value after *every* step. Tracing is the habit, not just the final answer.

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Common mistakes to watch for

- **Starting the trace from 0** instead of the value the variable was set to.
- **Adding when the program says take away**, or the other way round.
- **Writing only the final value** and skipping the steps in between.

Different paces

Support: Use programs of 2 steps with small numbers.
Stretch: Use two variables that change together, such as Asha's and Bala's toffees when one gives to the other.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Set score to 5. Add 3. Take away 2. What is score? (Answer: 6)

2. Set toffees to 12. Give away 4, then get 3. What is toffees? (Answer: 11)

Note in the journal: Beginning (needs help to use a variable to keep count); **Developing** (can use a variable to keep count with prompts or a partner); **Secure** (can use a variable to keep count and trace a number step by step alone); **Extending** (can find the start by undoing steps and explain why).

Answer key

1. 37
2. 35, then 55
3. Rs 35
4. a) 38
5. c) same: 12 each
Build it: Example (add 2, take 1): 2, 4, 3, 5
Challenge: Asha 13, Bala 7

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TEACHER NOTES · CHAPTER 13 · LEVEL B2

13

Time Goes On

Coder skill: Sequences and loops in time

Goes with NCERT Maths Mela, Class 3, chapter 13



At a glance	
Students will	put events in time order, use the week loop and read a clock.
Key words	**sequence** (things in order), **loop** (comes back to the start)
Materials	Day-of-the-week cards, a calendar page, the kit clock
Time	5 periods of 40 minutes
Link	Teach alongside chapter 13 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *sequence*, *loop*. Work through the example *What day is it 10 days after Monday?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The day circle	Run the activity below. Move between groups and note who can explain the idea.	Walk the day circle to find days before and after, and spot that the days repeat every 7.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Write a loop that says the days of the week. Model the first card or step, then let pairs finish.	How many times must the loop repeat to say every day once?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

Hands-on activity: The day circle

1. Seven children hold day cards and stand in a circle in order, Monday to Sunday.
2. A 'counter' child starts at a day and walks forward one card per day: 'What day is 3 days after Friday?'
3. Ask: 'What happens after 7 steps?' (back to the same day). So 10 days after Monday is the same as 3 days after Monday.
4. Write the loop: *repeat 7 times: say the next day*.

Tip: The circle shows why the week 'wraps round'. Link it to the alphabet circle from chapter 1.

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Common mistakes to watch for

- **Counting the starting day as day 1.** The first step lands on the next day.
- **Forgetting that the week wraps round** after Sunday.
- **Mixing up 'days after' and 'days before'**; walk backwards for 'before'.

Different paces

Support: Count on with a calendar page before using the circle.
Stretch: What day is it 30 days after Monday? (Wednesday: 30 days is 4 weeks and 2 days.)

Exit check and what to note

Two quick questions at the end of the chapter:

1. What day is 8 days after Tuesday? (Answer: Wednesday)

2. A loop says each day of the week once. How many times must it repeat? (Answer: 7)

Note in the journal: Beginning (needs help to put events in time order); **Developing** (can put events in time order with prompts or a partner); **Secure** (can put events in time order and use the week loop alone); **Extending** (can read a clock and explain why).

Answer key

1. 1 wake up, 2 go to school, 3 lunch, 4 dinner
2. b) Monday
3. 24th
4. A 3:00, B 6:30, C 9:00
5. b) games
Build it: Repeat 7 times
Challenge: Thursday (14 days = 2 loops, then 1 more day)

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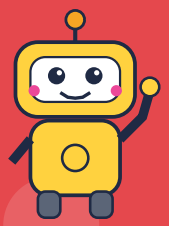
14

TEACHER NOTES · CHAPTER 14 · LEVEL B1-B2

The Surajkund Fair

Coder skill: Grids and robot routes

Goes with NCERT Maths Mela, Class 3, chapter 14



At a glance	
Students will	name squares on a grid, follow and write a robot program and find a bug in a route.
Key words	**grid** (rows and columns of squares), **program** (instructions a robot follows), **route** (the path you take)
Materials	Floor grid mat (or a 4 x 5 grid drawn on the floor), arrow cards, stall and wall cards
Time	5 periods of 40 minutes
Link	Teach alongside chapter 14 of the NCERT Class 3 maths book, *Maths Mela*.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *grid*, *program*, *route*. Work through the example *Take Robo to the toy stall* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Human robot routes	Run the activity below. Move between groups and note who can explain the idea.	Predict and walk arrow programs on the floor grid, fix routes that hit walls, and plan routes for others.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Plan Robo's trip around the fair. Start at A1, visit the sweet stall S, then the kite stall K, without crossing the grey walls. Model the first card or step, then let pairs finish.	Draw your arrows in order. Count them. Can a friend find a shorter route?
5	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Human robot routes

1. Put 'Robo' (a child) on A1 and a stall card on C3. Hold up arrow cards: up, up, right, right.
2. The class predicts where Robo will stop; Robo walks one square per arrow while the class reads them aloud.
3. Add a wall card on B3. Ask pairs to change the route so Robo avoids the wall, and test it on the mat.
4. Pairs plan a trip to two stalls with arrow cards, then swap routes with another pair and test them.

Tip: Arrows here are map directions (up means towards the top of the grid), not turns. Keep Robo facing the same way.

Common mistakes to watch for

- **Counting the start square as a step.** Ask: 'Did Robo move yet?'
- **Mixing up left and right.** Use map directions and keep Robo facing the top of the grid.
- **Skipping a square when walking the route.** Have the class count each move aloud.

Different paces

Support: Use a 3 x 3 grid and only up and right arrows.
Stretch: Shorten a route with *repeat*, for example *repeat 3 times: right*, and find the shortest route.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Robo is on A1. Write the arrows to reach C2. (Answer: for example right, right, up (in any order: 2 rights and 1 up))

2. With a wall on B2, the program up, right, right fails. Which arrow is the bug? (Answer: the 2nd arrow (it moves Robo into B2))

Note in the journal: Beginning (needs help to name squares on a grid); **Developing** (can name squares on a grid with prompts or a partner); **Secure** (can name squares on a grid and follow and write a robot program alone); **Extending** (can find a bug in a route and explain why).

Answer key

1. a) S
2. 7 arrows: 4 right and 3 up, in any order
3. b) the 2nd arrow hits the wall at B2
4. Repeat 4 times: right
5. a) D2
Build it: Example: up, right, right, right (S), right, up, up (K)
Challenge: Repeat 3 times: right, up

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Observation journal

Class 3 ____ Section ____ Term ____ Chapters _____. Circle one level per skill: **B**eginning, **D**eveloping, **S**ecure, **E**xtending. You may photocopy this page.

Student	Following steps	Patterns	Breaking down	Logic and rules	Notes
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	

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Term summary for the report card

One copy per child each term. Fill it from the observation journal and the child's self-checks. You may photocopy this page.

Name _____ Class _____ Section _____ Term _____

Area	Level (circle one)	Teacher comment
Computational thinking (overall)	B D S E	
Following and writing steps	B D S E	
Patterns and loops	B D S E	
Logic, rules and decisions	B D S E	
Coding on paper ('build it' tasks)	B D S E	

Writing comments

Write one strength and one next step, in words a parent understands. For example: *Explains steps clearly and fixes bugs in routes; next step: spotting the core of a pattern.*

Class teacher _____ Date _____

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